Neural Network Field Guide

42 architectures and training schemes, from the perceptron (1958) to diffusion LLMs (2025).

Each diagram draws a network in miniature, from input at the bottom to output at the top. In most panels, a circle is one neuron; in sequence models, a column is a token and a circle its whole vector at one layer. A square is larger: an image patch, a matrix-valued state, a whole network (in gray). Color and mark both show a cell's role, so diagrams read without color. Faded parts play little or no role in the step shown. Parts repeated over positions, windows, copies, or passes share weights; layers, experts, encoders, and decoders drawn alike do not. Sections group by a defining trait: wiring, dynamics, training, cost, or purpose. Years and training methods follow landmark papers, not always the first; many can be trained other ways.

Input

- Input unit
- Image patch
- Noisy input
- Random noise
- Visible unit (in and out)
- Masked token

Hidden

- Hidden unit
- Probabilistic hidden
- Capsule (vector output)
- Router

State

- Recurrent unit
- Memory cell
- Matrix-valued state

Mixing

- Attention
- Convolution or pooling

Output

- Output unit
- Reconstruction of input
- Velocity

Network

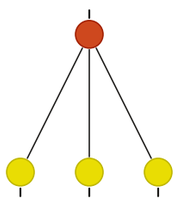
- A whole network

Links

- Connection
- Skip connection
- Data passed on or fed back
- Gate, acting at the dot
- Cross-attention, comparison
- Learnable function

FEED-FORWARD

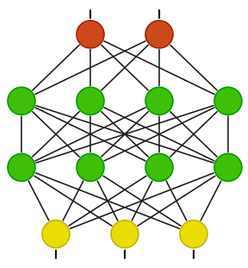
Perceptron



Weighted sum, then a threshold

1958 · supervised

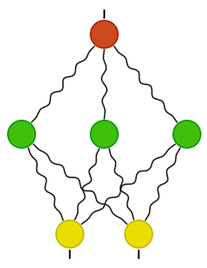
MLP



Stacked dense layers

1986 · supervised

KAN

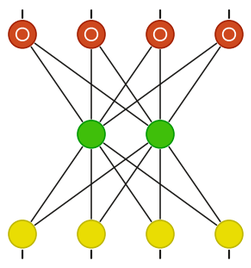


Learnable functions on edges

2024 · supervised

AUTOENCODERS

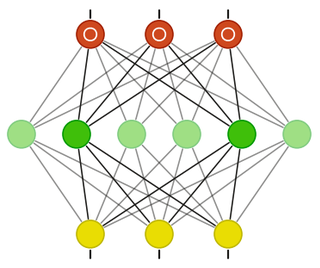
Autoencoder



Squeeze, then reconstruct

1986 · reconstruct

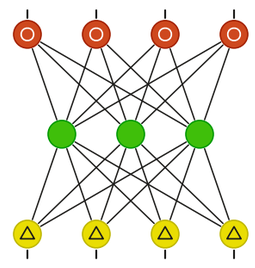
Sparse autoencoder



Wide code, few active units

2006 · reconstruct

Denoising autoencoder

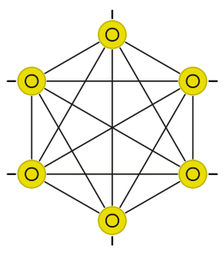


Clean up a noisy input

2008 · denoise

ENERGY-BASED AND SELF-ORGANIZING

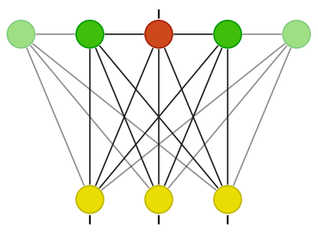
Hopfield network



Settles toward a stored pattern

1982 · hebbian

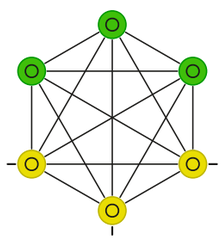
Kohonen map



Winner and neighbors move

1982 · competitive

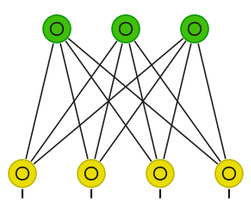
Boltzmann machine



Stochastic, fully connected

1985 · likelihood

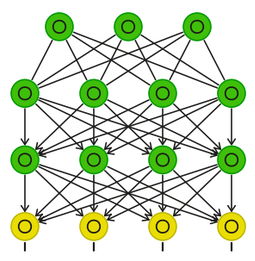
RBM



No links within a layer

1986 · likelihood

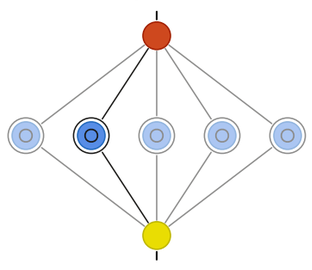
Deep belief network



Stacked RBMs, trained greedily

2006 · likelihood

Modern Hopfield network

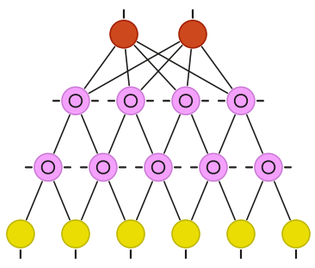


Attention as memory lookup

2020 · supervised

CONVOLUTIONAL

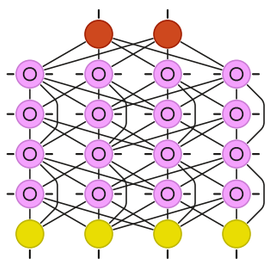
CNN



Local filters, shared weights

1989 · supervised

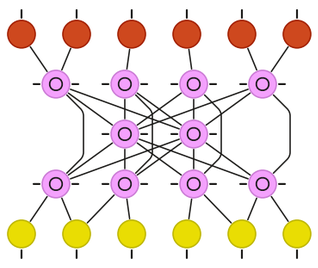
ResNet



Blocks add to a skip path

2015 · supervised

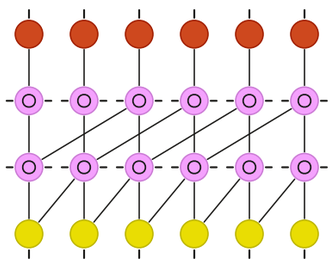
U-Net



Shrink, expand, skip across

2015 · supervised

WaveNet

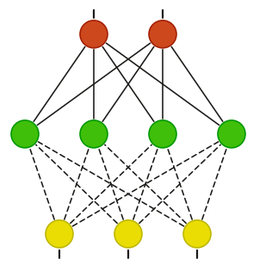


Dilated causal convolutions

2016 · next token

SIMILARITY

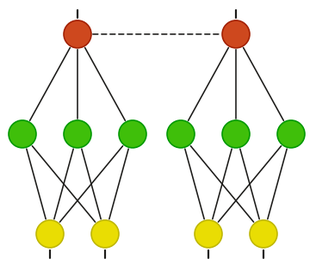
Radial basis network



Distance to stored centers

1988 · supervised

Siamese network

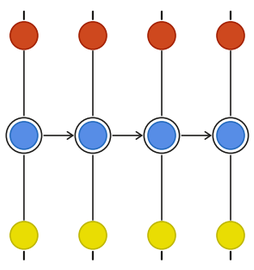


Twin towers, shared weights

1993 · contrastive

RECURRENT AND MEMORY

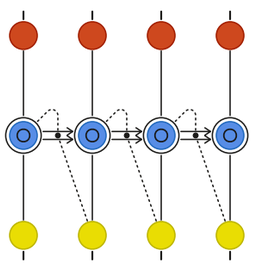
RNN



Hidden state passed along

1990 · next token

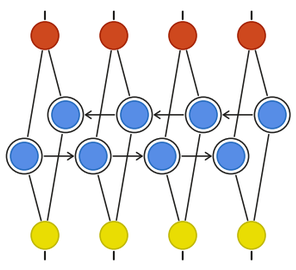
LSTM



Two states; forget gate from 1999

1997 · next token

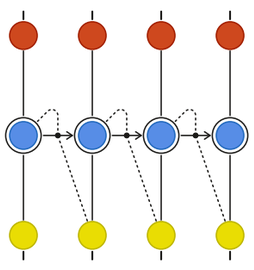
Bidirectional RNN



Reads the sequence both ways

1997 · supervised

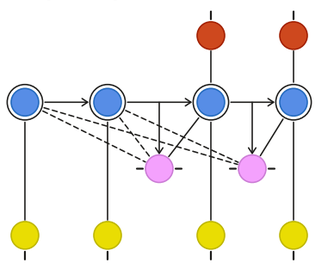
GRU



One state, two gates

2014 · next token

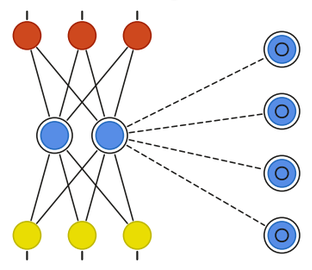
Seq2seq with attention



Learns to align and translate

2014 · next token

Neural Turing machine

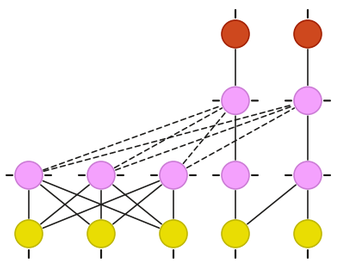


Controller reads, writes memory

2014 · supervised

ATTENTION

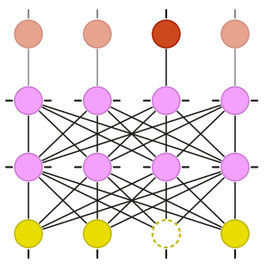
Transformer



Encoder, decoder, cross-attention

2017 · next token

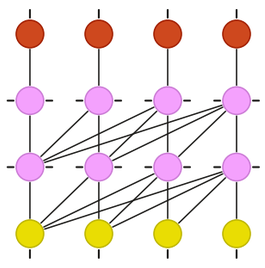
BERT



Every token sees every token

2018 · masked token

GPT-style LLM



Sees itself and earlier tokens

2018 · next token

ViT

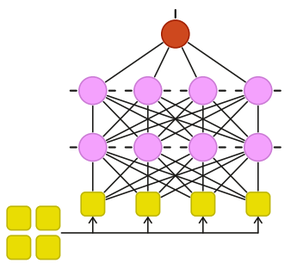
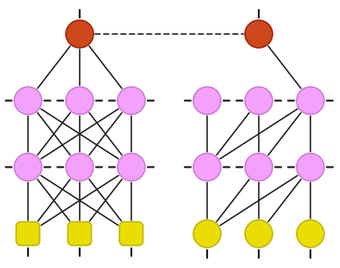


Image patches become tokens

2020 · supervised

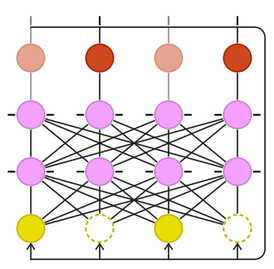
CLIP



Match images to captions

2021 · contrastive

Masked diffusion LLM

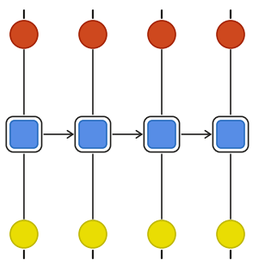


Unmask in parallel, step by step

2025 · masked token

LINEAR-TIME MODELS AND HYBRIDS

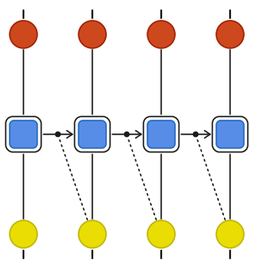
Linear attention



Softmax-free attention as an RNN

2020 · next token

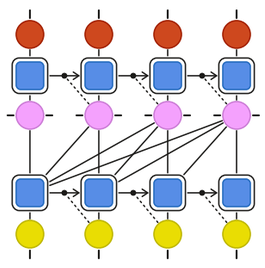
Mamba



Selective state, linear time

2023 · next token

Mamba–attention hybrid

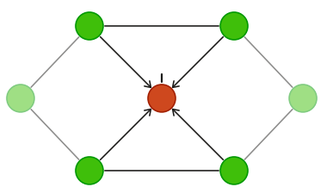


Mostly recurrent, some attention

2024 · next token

ROUTING AND GRAPHS

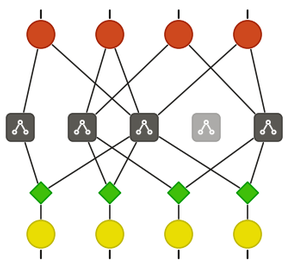
Graph neural network



Nodes pool neighbor messages

2005 · supervised

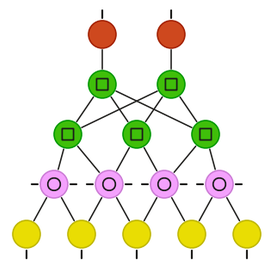
Sparse MoE



Each token → its top-*k* experts

2017 · next token

Capsule network

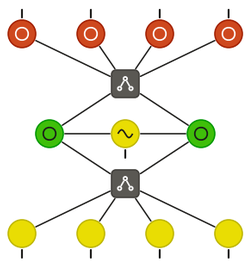


Vectors routed by agreement

2017 · supervised

GENERATIVE FRAMEWORKS

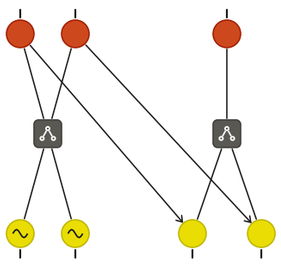
Variational autoencoder



Sample a code, then decode

2013 · variational

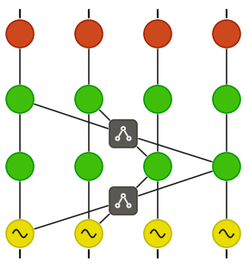
GAN



Generator vs. discriminator

2014 · adversarial

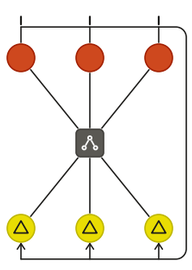
Normalizing flow



Invertible map from noise to data

2014 · likelihood

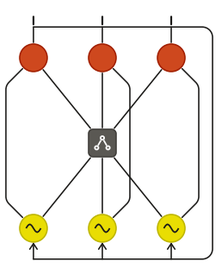
Diffusion model



One denoiser, run *T* times

2015 · variational

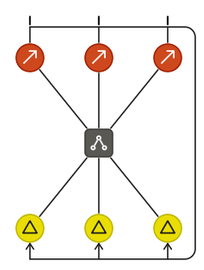
Neural ODE



A flow with continuous depth

2018 · likelihood

Flow matching



Learns a velocity, noise to data

2022 · flow matching

Inspired by The Neural Network Zoo by Fjodor van Veen (Asimov Institute, 2016; updated with Stefan Leijnen, 2019). www.asimovinstitute.org/neural-network-zoo

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